

Key: The Ground Moves Under the Data

84-355 Democracy's Data | Precinct geography | Instructor copy — do not distribute

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All code is given to students. Grade the interpretation.

Structure. The brief (`precinct-geography-brief.Rmd`) is now a chapter, “The Ground Moves Under the Data”: it is organised by argument under prose headings and carries four numbered D3 figures rather than numbered steps — Figure 1, Houston County’s precincts in 2020 and 2024; Figure 2, disaggregate-then-reaggregate in one sketch; Figure 3, the one piece of Houston ground that changed precinct; Figure 4, where land and people give different answers across Georgia. The lab (`precinct-geography.Rmd`) is unchanged and still runs Steps 1–10 with the code exposed. **Students should read the brief first. Step numbers below refer to the lab.** The chapter covers the same ground in the same order, so each lab Step has a heading facing it: Step 1 = “Nobody in Atlanta drew these lines”; Step 2 = “Sixteen precincts you can name” (Figure 1); Step 3 = “The repair that deletes the wrong precincts”; Steps 4 and 5 = “A unit that both maps contain” (Figure 2); Step 6 = “Land or people”, worked out in “One precinct, taken apart” (Figure 3) and scaled up in “How bad it gets elsewhere” (Figure 4); Steps 7, 8 and 9 = “Carrying the votes, and checking that they arrived”; Step 10 = “What the crosswalk licenses”.

The worked example is Houston County, and the statewide numbers stay. Every step is illustrated in one county you can talk about from memory, then checked against Georgia so nobody thinks the county was chosen for its answer. Where the two differ, the difference is the lesson — most sharply in Step 6, where Houston is a *stronger* case than the state, and in Option B, where it is a much weaker one.

This is a methods session, and it is the only one in the course that teaches a procedure students could be asked to perform. Everything else asks them to read data somebody else assembled. Here they learn a thing redistricting analysts, expert witnesses and election reporters actually do, and why it is contested.

Every number below is computed from the built files at knit time.

The session in one sentence

Precinct boundaries move between elections, so comparing precincts across years means putting both years on one map — and the reasonable ways of doing that disagree with each other about something as basic as whether a precinct was split.

The findings

```
rbind(
  HOUSTON = c(name_survival_pct = sv(h20, h24),
    precincts_2020 = length(h20),
    blocks = nrow(baH),
    population = sum(baH$pop),
    areal_split_pct = sp(arH),
    pop_split_pct = sp(poH),
    pairs_over_10pts = sum(abs(cpH$diff) > 0.10),
    worst_gap = round(max(abs(cpH$diff)), 3)),
  GEORGIA = c(sv(p20, p24), length(p20), nrow(ba), sum(ba$pop, na.rm = TRUE),
    sp(ar), sp(po), sum(abs(cp$diff) > 0.10), round(max(abs(cp$diff)), 3)))
```

```
#>      name_survival_pct precincts_2020 blocks population areal_split_pct
#> HOUSTON              81.2           16   3269     163633           93.8
#> GEORGIA              84.5          2651 232717    10711908           78.3
#>      pop_split_pct pairs_over_10pts worst_gap
#> HOUSTON           6.2              2    0.127
#> GEORGIA           18.5             324    0.890
```

- **81.2% of Houston's precinct names survive 2020→2024; 84.5% of Georgia's do.** Statewide, precinct *codes* do worse still — about 75%. Two keys, two answers.
- **Houston: 3,269 blocks, 163,633 people.** Georgia: 232,717 blocks, 10,711,908 — the census count exactly.
- **Areal weighting calls 93.8% of Houston's precincts split; population weighting calls 6.2%.** Statewide: 78.3% and 18.5%. **The county's contrast is the sharper one** — that is why the example moved there.
- **11.8% of Houston's weight pairs differ by more than ten points, worst 0.127.** Statewide 9.5%, worst 0.890. **Do not quote the statewide extreme as Houston's** — see the note under Option B.
- The carry loses **22 votes of 13,821,357.**

The Houston inventory, which is the new centre of the session

```
aH <- inC(a); aH <- aH[order(aH$from_2020), ]
data.frame(precinct_2020 = nm(aH$from_2020), became_2024 = nm(aH$to_2024))
```

```
#>      precinct_2020      became_2024
#> 1          ANNX          ANNX
#> 2           BMS          BMS
#> 3          CENT          CENT
#> 4          CGTC          CGTC
#> 5          FMMS          FMMS
#> 6          HAFS          HAFS
#> 7          HCTC          VFW
#> 8          HEFS          HEFS
#> 9          HHPC          HHPC
#> 10         MCMS          MCMS
```

```
#> 11      NSES NORTH HOUSTON SPORT COMPLEX
#> 12      RECR                WELLSTON CENTER
#> 13      ROZR                ROZR
#> 14      TMS                 TMS
#> 15      TWPK                TWPK
#> 16      VHS                 VHS
```

```
setdiff(nm(h24all), nm(h24))      # the 2024 precinct no interior point sees
```

```
#> [1] "PEC"
```

16 lines, and they tell three different stories.

1. **13 precincts are unchanged** in name and, as the crosswalk confirms, in ground.
2. **3 are renames**: HCTC→VFW, NSES→NORTH HOUSTON SPORT COMPLEX, RECR→WELLSTON CENTER. Population weighting sends **100%** of each old precinct's people to the new name; areal overlap is above 99.7%. **The polling place moved and the boundary did not.**
3. **1 is a real split**: ROZR shed its western half to a brand-new precinct, PEC.

```
po[po$from_2020 %in% ren, ]      # the renames: weight is exactly 1
```

```
#>      from_2020      to_2024 weight  pop
#> 2445 HOUSTON|HCTC      HOUSTON|VFW      1  9846
#> 2449 HOUSTON|NSES HOUSTON|NORTH HOUSTON SPORT COMPLEX      1  3517
#> 2450 HOUSTON|RECR      HOUSTON|WELLSTON CENTER      1 17798
```

```
po[po$from_2020 %in% spl, ]      # the one real split
```

```
#>      from_2020      to_2024  weight  pop
#> 2451 HOUSTON|ROZR HOUSTON|ROZR 0.571018 11908
#> 2452 HOUSTON|ROZR HOUSTON|PEC  0.428982  8946
```

This is the single biggest teaching win of the county version, and it is worth protecting time for. The statewide statistic — 84.5% of names survive — silently merges two completely different events. In Houston you can separate them by name, out loud, in ninety seconds: **3 of the 4 changes are clerical and 1 is real**. Ask the room which of them a name-matched comparison drops. The answer is *the 3 that did not matter* — the rule deletes the harmless changes and keeps the one that would have broken the comparison.

The three things to get said

1. **An identifier that is not stable is not an identifier.** Step 2 gives two different answers depending on whether you key on name or on code, and both are computed correctly. **Make somebody say why that is worse than either number alone**: if the two keys disagreed only occasionally you would call it noise, but they drift independently, which means neither is tracking the thing you care about — a piece of ground. Houston makes it concrete: 81.2% of names survived, and only 1 of the county's 16 precincts changed its boundary at all.

2. The dropped precincts are not missing at random (Step 3). This is the step students will skate over, and it is the one with real consequences. Statewide the argument is that counties split precincts where population grew, so dropping non-matching precincts deletes the fastest-growing places. **In Houston the mechanism is visible and it points the other way:** name-matching drops the three renames and keeps ROZR, the only precinct whose ground moved. Either way, **name-matching is not a filter on boundary change at all** — it is a filter on clerical churn. Ask what a turnout-change map looks like when the filter is uncorrelated with the thing it is supposed to control for.

3. Step 6 is the session. Two defensible methods, computed on the same data, disagreeing about whether 93.8% or 6.2% of a county’s precincts were split. Neither is a bug. Area answers “how much land moved”; population answers “how many people moved.” **The lesson is not which to pick — it is that a methodological choice this invisible moves the answer this much.**

New this version, and worth two minutes: **most of what area calls a split is a sliver.**

```
c(houston_area = sp(arH), houston_area_1pct_floor = sp(arH, 0.01),
  houston_pop = sp(poH), houston_pop_1pct_floor = sp(poH, 0.01),
  georgia_area = sp(ar), georgia_area_1pct_floor = sp(ar, 0.01),
  georgia_pop = sp(po), georgia_pop_1pct_floor = sp(po, 0.01))
```

```
#>          houston_area houston_area_1pct_floor          houston_pop
#>          93.8          6.2          6.2
#> houston_pop_1pct_floor          georgia_area georgia_area_1pct_floor
#>          6.2          78.3          18.7
#>          georgia_pop georgia_pop_1pct_floor
#>          18.5          14.7
```

Ignore any fragment smaller than 1% of a precinct and Houston’s areal figure falls from 93.8% to 6.2%, exactly matching population. Statewide the same correction takes area from 78.3% to 18.7% and population from 18.5% to 14.7%; the 4-point residue is genuine empty land. **This links Step 5 to Step 6** — the slivers Step 5 says the interior-point method avoids are the same slivers inflating area’s answer in Step 6.

The line for the board: the map is not the ground, and the label is not the map.

Walking the steps

Step	What it does	Min	Watch for
1 Who owns a boundary	The state’s own “best fit” language	6	Read the quotation aloud. Students assume the state draws precincts. It does not. Houston’s board does.
2 How much moved	81.2% of Houston’s names, 84.5% of Georgia’s	7	The two-key disagreement is the finding, not a nuisance.
2b The inventory	16 lines: 13 same, 3 renamed, 1 split	8	New, and the best thing in the lab. Do it by name, out loud.

Step	What it does	Min	Watch for
3 Why you cannot drop them	Non-random missingness	7	Houston drops the three harmless ones and keeps the real one.
4 The recipe	Target, disaggregate, reaggregate	5	Houston: 3,269 blocks, 204 per precinct.
5 How a block is assigned	Interior point, not intersection	6	PEC is invisible to a precinct-level interior point. That is why blocks.
6 Two weightings	93.8% vs 6.2% in Houston	12	The pivot. Take a vote before revealing the second number.
7 Carry and check	22 votes of 13,821,357	8	The 9.8% failure story lives here. Do not rush it.
8 Was the shortcut legitimate	2,355 of 2,355 identical	4	Check <i>before</i> relying, not after.
9 What you can say	Including what you cannot	6	Population is not voters.

Step 6 deserves the prediction. Ask the room: *areal weighting says 94% of Houston's precincts are split by the boundary change — what will population weighting say?* Almost everyone predicts a similar number. It is 6.2% — **1 precinct** — and the reason (most splits are slivers or empty land) is obvious in hindsight and almost never guessed. Houston makes the reveal sharper than the statewide version did, because “1 precinct out of 16” is a fact about a place rather than a percentage.

If you are short on time, protect 2b, 6 and 7. The inventory, the two weightings, and the join that lost a tenth of the state are the argument.

The mistake in Step 7, and how to teach it

Tell this story, because it happened while building the lab.

The first version attached votes to the map by matching precinct names. It reached 90.4% and lost **489,876 votes — 9.8% of Georgia** — concentrated in Chatham (92 precincts) and DeKalb (48), where the returns and the shapefile spell precincts differently.

The fixes attempted, in order:

1. Strip a leading ballot-order number: "01 BETHLEHEM COMMUNITY CENTER" → "BETHLEHEM COMMUNITY CENTER". Loss fell from 11.1% to 9.8%.
2. Allow a trailing letter on the code: "01C FIRST PRESBYTERIAN CHURCH". Loss did not move at all.

Neither was the answer, and the third attempt was not another regex. The precinct shapefile already carried the vote totals in its own attribute table, attached to the geometry. There was never a join to make.

The pedagogy: **ask the class what they would have tried third.** Most will propose fuzzy matching or a manual alias table — both reasonable, both more work, both unnecessary. The instinct to improve a join is strong enough that it crowds out the question of whether the join is needed.

Then note what made it safe to abandon: Step 8 shows the two state sources agree on every precinct the join reached.

```
ts

#>      scope matched identical differing shapefile_precincts
#> 1 Georgia      2355      2355          0                2684
#> 2 HOUSTON       16       16          0                16

round(100 * ts$matched / ts$shapefile_precincts, 1)    # what the join reached

#> [1]  87.7 100.0
```

The shortcut was verified before it was relied on, which is the difference between a shortcut and a guess. And note the second line: the join reached 87.7% of Georgia’s precincts and **100% of Houston’s** — the data was never wrong, the labels were.

A note on a number you may remember. Earlier drafts of this key quoted “2,401 of 2,401” for the two-source check. That figure came from a normalisation of the returns that is no longer in the pipeline. Recomputed from the files as they now stand, the plain COUNTY|PRECINCT join reaches 2,355 precincts and **agreement is still exact on every one of them**, which is the part that matters. The chunk above is the live number; use it.

Handling the room

This lab is more technical than the rest of the course and a few students will decide it is “the coding one” and disengage. Three things help:

- **Nobody writes code here either.** Say so. The procedure is conceptual — target, disaggregate, reaggregate — and the code is given.
- **Anchor it in a place, not a statistic.** The county version exists for this. “Warner Robins” and “the precinct at the tech college” beat “78.3% of precincts” every time; the statewide figure is there to stop anyone concluding that Houston is weird.
- **The disagreement in Step 6 is the interesting part**, not the machinery. If the room only takes away that two reasonable methods said 93.8% and 6.2% about the same 16 precincts, that is a successful session.

Option A — Use the areal crosswalk instead

Scope: Houston, 17 precincts. Only one of them moves at all, which is the point.

Verified output

```
big <- cpH[order(-abs(cpH$diff)), c("from_2020", "to_2024", "weight_area", "weight_pop", "diff")]
big[, 3:5] <- round(big[, 3:5], 3)
head(big, 5)
```

```
#>           from_2020           to_2024 weight_area weight_pop
#> 260 HOUSTON|ROZR           HOUSTON|ROZR      0.444      0.571
#> 261 HOUSTON|ROZR           HOUSTON|PEC      0.556      0.429
#> 1459 HOUSTON|NSES HOUSTON|NORTH HOUSTON SPORT COMPLEX      0.997      1.000
#> 1715 HOUSTON|HCTC           HOUSTON|VFW      0.999      1.000
#> 1726 HOUSTON|ANNX           HOUSTON|ANNX      0.999      1.000
#>           diff
#> 260      0.127
#> 261     -0.127
#> 1459      0.003
#> 1715      0.001
#> 1726      0.001
```

```
wm      # ballots, both ways, for the one precinct that actually splits
```

```
#>  target  w_area  w_pop  pop ballots_area ballots_pop ballots_total
#> 1   ROZR 0.444468 0.571018 11908      4460      5730      10035
#> 2    PEC 0.555529 0.428982  8946      5575      4305      10035
```

3/4: Swaps the file, recomputes, and reports that 15 of Houston’s 16 precincts do not move at all.

4/4: Identifies ROZR as the only precinct where the choice matters, quantifies it (**1,270 ballots** out of 10,035 cast), and explains *why*: the western half is large and thinly settled, so area over-weights it and population does not.

4/4, best: Generalises from the one case — area and population agree wherever population density is uniform and diverge wherever it is not, so **the choice of method is effectively a choice about how much sparsely settled ground counts**. A student who then checks that claim against the statewide scatter in the brief (Figure 4, “How bad it gets elsewhere”), and notices that the off-diagonal points are the low-density ones, has done the whole exercise.

Option B — Find the worst disagreement in Georgia

Scope is statewide, and the lab now says so explicitly. Houston’s worst disagreement is only 0.127 — the ROZR split — so a student who searches only the county will not find anything dramatic and should not be marked down for reporting that. **The 0.890 case does not exist in Houston.** If a student’s answer quotes 0.890 as a Houston number, that is the error to correct.

Verified output

```
w <- cp[which.max(abs(cp$diff)), ] # cp already has "NA/NA" dropped
data.frame(from = w$from_2020, to = w$to_2024,
           area = round(w$weight_area, 3), pop = round(w$weight_pop, 3),
           gap = round(w$diff, 3))
```

```
#>               from               to area pop gap
#> 1 CHATHAM|ELI WHITNEY COMPLEX CHATHAM|ELI WHITNEY COMPLEX 0.11    1 0.89
```

```
# for contrast, the same query inside the county
cpH[which.max(abs(cpH$diff)), ]
```

```
#>      from_2020      to_2024 weight_area weight_pop    diff
#> 260 HOUSTON|ROZR HOUSTON|ROZR    0.444468    0.571018 0.12655
```

Watch for the junk row. A student who forgets the "NA|NA" guard will report 0.894 from the pseudo-precinct instead of 0.890 from ELI WHITNEY COMPLEX. Noticing the guard is itself worth credit; the lab and the brief both print it.

3/4: Reports the pair and the size of the gap.

4/4: Works out what the gap means concretely — one method sends about 11% of that precinct’s votes to the target, the other sends about 100%. Says which is more likely correct and why.

4/4, best: Recognises that a single pair cannot be adjudicated from the table, and that settling it needs a map or local knowledge — then says what they would look for (a river, an industrial tract, a park). **Knowing that the data cannot answer it is the right answer.** The strongest version notices the asymmetry the county example sets up: *in Houston we could adjudicate, because somebody in the room knows the ground; in Chatham nobody in the room does, and the method has no way to tell you that it is out of its depth.*

Option C — Turnout, honestly

Scope: Houston. No fully verified output; grade the reasoning and the comparison.

3/4: Computes turnout change both ways and reports that the numbers differ.

4/4: Quantifies the difference and names the three precincts the naive method drops (HCTC, NSES, RECR), noting that the naive method is not merely noisier but **structurally wrong** — it deletes precincts for a clerical reason unrelated to whether the ground moved.

4/4, best: Notices that even the crosswalked version has a problem the lab warned about: **2020 turnout carried on 2020 population weights, compared against 2024 turnout, attributes to “turnout change” some of what is really population change.** In a county that split a precinct precisely because it was growing, that is not a hypothetical. A student who sees that has understood the limitation better than the method does.

Grading

Four-point scale from the syllabus.

- **Never penalise code.** It is given.
- **The move that earns a 4** is treating the weighting choice as a substantive decision with consequences, not a technical detail.

- **Also a 4:** separating “the name changed” from “the boundary changed.” Houston hands them the distinction; using it on a number they did not compute themselves — the statewide 84.5% — is the transfer worth rewarding.
- **A 2** is “the boundaries changed so the data is unreliable” — true, useless, and the thing the whole lab is designed to move past.
- **Also a 2:** reporting the crosswalked numbers as though they were counted.
- **Also a 2:** quoting a Houston illustration as though it were a statewide finding, or the reverse. The documents are built to make the scope explicit; ignoring it is the failure mode the county version introduces.
- **Reward anyone who asks for an error bar.** There isn’t one, and noticing its absence is the most sophisticated available response.

Anticipated questions

- “*Why Houston?*” — Because it is small enough to check by hand and because the instructor knows the ground. Then say what the statewide columns are for: if the county were doing the work, they would not be there.
- “*Is Houston typical?*” — On name survival, almost exactly (81.2% against 84.5%). On the area-versus-population split it is more extreme, and the honest answer is that this is partly a small-county artefact: with 16 precincts, 1 real split is 6.2%.
- “*Why not just use counties?*” — You can, and it is stable. You also lose all the resolution that makes precinct data worth having. This is a real position some analysts hold; say so.
- “*Couldn’t we weight by registered voters?*” — Yes, and it would be better. Nobody publishes registered voters by census block. Worth naming as the gap.
- “*Is this what expert witnesses do?*” — Yes. Effectively this procedure, with more care about which small unit to use.
- “*How would I know if a precinct changed?*” — By noticing that it did. There is no notification, no changelog, and no requirement to publish one. That answer usually lands harder than anything else in the session.

If you want to check a Houston number against a third source

Houston County publishes its own precinct-level results at <https://www.houstoncountygga.gov/residents/election-results.cms>. Nothing in the lab is built on it and no chunk fetches it. It is worth knowing about for two reasons: it is a genuinely independent copy of the same totals, so it settles any question about a Houston figure on the spot; and it is a live example of the Step 8 point that three offices publish the same numbers separately and the only thing that ever disagrees is the labels.

Rebuilding the figures

`build-houston-figures.R` in this folder reads the shapefiles and block assignments in `../ga-precinct-returns/data/` and writes the small coordinate tables the maps draw from (`fig_houston*.csv`, `fig_rozr*.csv`, `fig_twosource.csv`). **Rerun it after any rebuild of the crosswalks**, then reknit all three documents; every number in them is computed at knit time, so a stale figure file is the only way they can go out of date.

Connections

- **census-geography** — the nesting spine, and why precincts sit off it. This lab is that lab’s abstract point made concrete and consequential.
- **ga-precinct-returns** and **rpv** — both use precinct data as though the unit were fixed. Run this first if you can.

- **redlining** — point-in-polygon, the same uniform-distribution assumption, and there we could not measure the error. Here we can.
- **apportionment** — the other lab where a technical rule with no obvious politics turns out to allocate something people care about.